



Unit 4 · Outcome 3 · Scientific Investigation

Assumptions & limitations

KK09

VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary — clean explanations, comparison tables, visual aids, exam traps, command words and worked answers.



Assumptions — the things you take as true

An assumption is a condition you accept as true without directly proving it, so that your investigation makes sense and your conclusion holds. Assumptions are usually unstated — which is exactly why examiners reward students who can drag them into the light.

◆ KEY INSIGHT

Analogy · the foundations you never see A conclusion is a house. Assumptions are the foundations underneath it — invisible once the house is built, but the house only stands if they hold. If a key assumption is false, the whole conclusion can subside, even though the walls looked fine.

For Dev's study to mean anything, several things must be true that Dev never actually checked:

- The two arrays are identical — same panel model, age, tilt and wiring — so the only difference between them is the shade.
- The inverter app logs accurately and consistently for both arrays.
- Apart from the tree's shadow, both arrays receive the same incoming sunlight each day.
- The two-week April window is representative of how the system normally behaves.
- Cloud cover affected both arrays equally on any given day.

Notice these aren't mistakes and they aren't equipment faults — they're premises. Dev's claim "the tree's shade reduces output" is only safe if those premises hold. Name the premise, and you've named the assumption.

✓ TIP

✓ How to phrase an assumption in the discussion Use the frame: "This investigation assumed that <X>, because the conclusion relies on <why it matters>." e.g. "It was assumed the two arrays were identical apart from shading, because any built-in difference between them would otherwise explain the output gap instead of the tree."

Limitations — what your study could not do

A limitation is a feature of your design, equipment or scope that restricts how confidently or how widely you can conclude. Crucially, a limitation is built in — it would still be there even if you ran the whole thing perfectly, with no slips.

◆ KEY INSIGHT

Analogy · the size of your window A limitation is the size of the window you're looking through. Dev studied one roof, one tree, one fortnight — a small window. Whatever Dev sees through it is real, but Dev cannot claim to know the whole landscape. The narrower the window, the narrower the conclusion you're entitled to.

Dev's study is genuinely useful, but it's fenced in. It can suggest the gum tree's shade lowered this array's output across these April days — and not much further. Limitations are what build that fence, and a strong discussion names them and says how each one narrows the conclusion.

"I might have misread the inverter screen" is not a limitation — it's an avoidable personal error. Limitations survive a perfectly-run experiment; mistakes don't. Section 04 drills this exact split, because examiners report it as the single most common error in U-O3 responses.

The three sources VCAA names

The wording of KK09 is a gift — it tells you exactly where to go looking. Strong students walk through all three categories like a checklist, which is also how the marks are usually split.

- Small sample / few replicates
- Narrow range of the independent variable
- Short time frame; single site or season
- A confounding variable you couldn't control
- No genuine control group
- Instrument resolution or range
- Reading to nearest 0.1 kWh only
- Daily totals, not continuous logging
- Observer judgement / subjectivity
- Too few repeat readings to average out noise
- A correlation, not proof of cause
- Can't extrapolate beyond the tested range
- Small $n \rightarrow$ low confidence in the trend
- Qualitative data hard to compare objectively
- Anomalies left in distort the mean
- Method: one roof / tree / fortnight; cloud not controlled
- Data: inverter rounds to 0.1 kWh; daily totals only
- Analysis: shade & output correlate, but April sun angle differs from summer — can't generalise to the year

Assumption vs limitation vs error

Three quick tests will place almost any statement:

- Assumption — is it a premise you're taking as true for the study to work? "We assume X is true."
- Limitation — would it still be there if you ran the study flawlessly? Then it's built into the design or scope. "The study can't..."
- Personal error / mistake — is it an avoidable slip a careful person could've prevented? Then it's an error, not a limitation. "I accidentally..."

From limitation to improvement

The poster's discussion — and most exam "evaluate" questions — want a limitation paired with a realistic improvement that actually addresses it. The trap is offering "be more careful," which fixes error, not the limitation. A real improvement changes the design, equipment or scope.

- **NOTE**

Link it back to the conclusion Top-band answers don't just list a limitation and a fix — they say what it does to the conclusion: "Because only April was sampled, the conclusion can't be extended to summer, when the sun is higher and the shadow shorter. Repeating across seasons would let the conclusion generalise across the year." That last clause — the effect on validity and scope — is where the marks live.

P

Writing it the way examiners reward

First, read the verb. The same scenario earns very different things depending on the command word:

For a limitations response, Point (name it + its category), Example (the specific detail from the study), Link (what it does to the conclusion + the fix). Type below and the meters track whether you've hit each move.

P A key methodology limitation is that diversity was sampled at only one site on a single afternoon.

E With three net sweeps at one location and one time, the sample is too small and too narrow to capture the variation that exists between sites, times of day and seasons.

L This means the result is not representative of the waterway as a whole, so the conclusion cannot be generalised beyond that spot and moment. Sampling several sites across multiple days would let the conclusion describe the waterway, not just one sweep of it.

Six exam traps

⚠ EXAM TRAPS

"I read the scale wrong" / "I forgot to start the timer" are personal errors. They're avoidable and don't belong in a limitations answer. A limitation survives a perfectly-run study. Writing "human error" earns nothing — it's vague and usually means a mistake anyway. Name a specific, structural limitation instead. "Small sample size" on its own is a list item, not an answer. Say what it does: reduces confidence in the trend, so the conclusion can't be generalised. "Be more careful" or "repeat to be accurate" addresses error, not the limitation. A real fix changes the design, equipment or scope — more sites, a finer instrument, a longer window. An assumption is a premise taken as true; a limitation is a constraint on the study. "We assumed the arrays were identical" is an assumption; "only one roof was tested" is a limitation. KK09 lists methodology, data generation and analysis. If a question asks for two limitations, drawing them from different categories shows range and de-risks repetition.

MULTIPLE CHOICE

Practice exam · 20 marks

- A premise accepted as true, without proof, that the comparison relies on.
- Distinguish from a limitation (a constraint) and an error (an avoidable slip).

One-screen cheat sheet

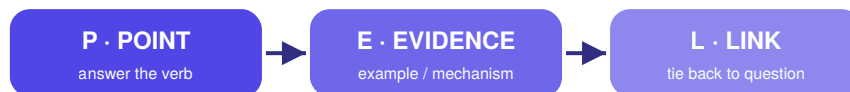
- Assumption — premise taken as true; the conclusion depends on it.
- Limitation — built-in constraint; restricts the conclusion.
- Error — avoidable slip; not a limitation.
- Methodology — sample size, range, time, site, confounders, no control.

- Data generation — resolution, subjectivity, too few repeats.
- Analysis — correlation $\neq$ cause, no extrapolation, small n.
- "We assume X." → assumption
- Survives a perfect run? → limitation
- A careful person avoids it? → error
- Name it + its category.
- Detail it from the study.
- Link to the conclusion (effect on validity/scope).
- Improve it by changing design/equipment/scope.
- More replicates / sites / seasons.
- Finer instrument; continuous logging.
- Control or measure the confounder.
- Never "be more careful" (that's error).
- Poster discussion section (SAC).
- Exam: evaluate-an-investigation items.
- Often paired with accuracy/validity (KK04).

EXAM TECHNIQUE

★ Worked answers — weak vs strong

The same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer shape — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

C — methodology limitation. Why it drops the mark: it confuses a stated premise taken as true with a constraint on the design. The student isn't saying the study couldn't control moisture — they're assuming it was equal.

✓ STRONG / MODEL ANSWER

B — an assumption. Why it's right: the word "assumes" and the structure "X was the same" mark it as a premise accepted as true so the comparison is fair — the definition of an assumption.

✗ WEAK ANSWER

An assumption is something you think is true and a limitation is a problem with the experiment." Why it caps at 0–1: "a problem with the experiment" is too loose — it doesn't separate a constraint from a mistake, and "something you think is true" misses that the study depends on it.

✓ STRONG / MODEL ANSWER

An assumption is a premise accepted as true without proof, that the method or conclusion relies on. A limitation is a feature of the design, equipment or scope that restricts how confidently or how widely the conclusion can be drawn — and would remain even if the study were run perfectly." Why it's full marks: defines both precisely and draws the line — depends-on vs restricts — without slipping into "error."

✗ WEAK ANSWER

Dev might have read the app wrong, and the experiment wasn't very long so it could have human error." Why it scores ~0–1: "read the app wrong" and "human error" are avoidable mistakes, not limitations. "Wasn't very long" gestures at a real point but has no category and no consequence.

✓ STRONG / MODEL ANSWER

Methodology: the study ran on one roof with one tree across a single two-week April window, so the sample is narrow and seasonal — the sun is lower in April than summer, so the result may not hold across the year. Data generation: the inverter logged only daily totals to the nearest 0.1 kWh, so small differences between arrays may fall below the resolution and within-day shading effects are invisible. Why it's full marks: two limitations, two different categories, each with a specific detail and a stated consequence.

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Worked good/weak answers illustrate exam technique — always check against current VCAA marking advice.